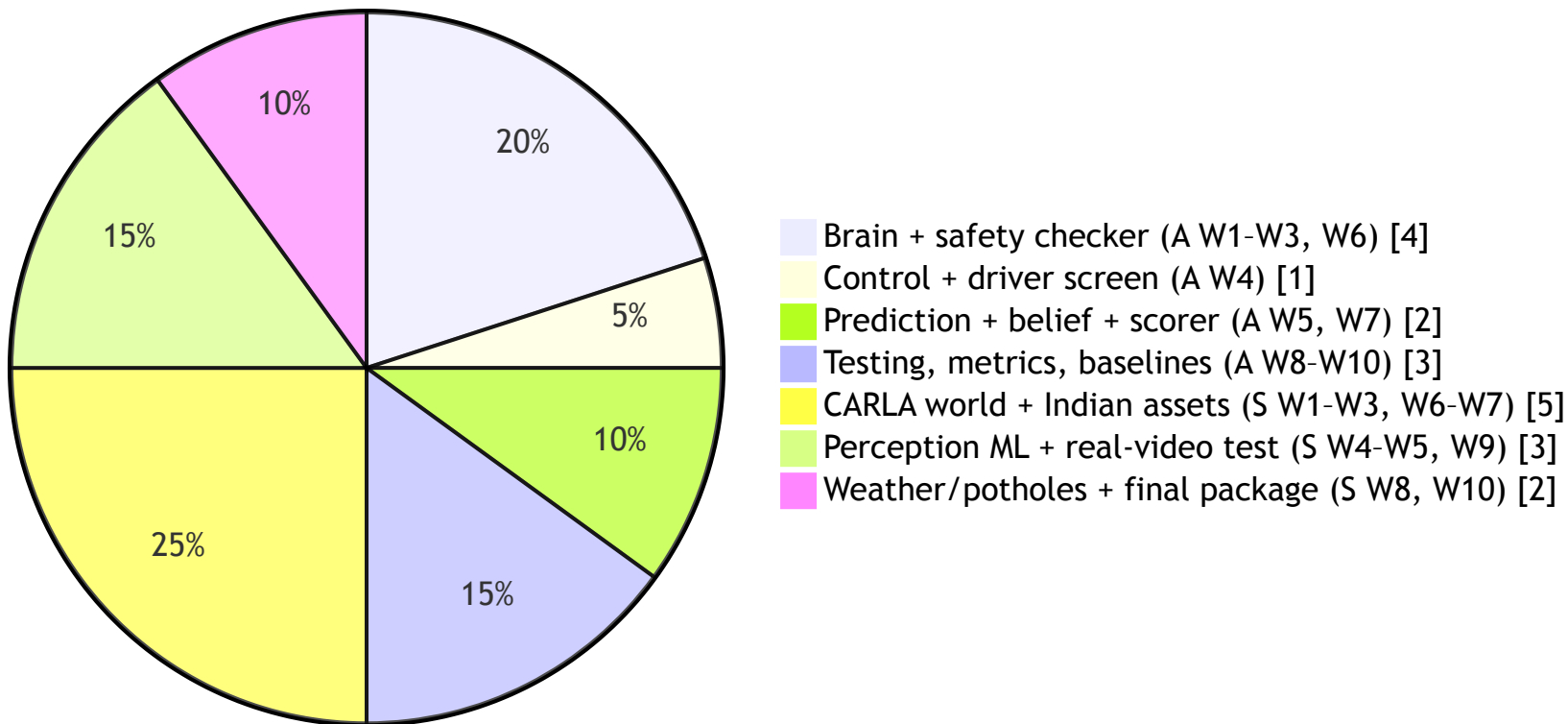


## D7 • Competitor comparison + cost

Pie = planned effort split (a plan, not a measurement, not a cost pie). Competitor table and sourced cash cost on the next page.

Planned effort split — 20 builder-weeks (2 builders × 10 weeks)



# D7 · Competitor comparison + cost — exact spec for redrawing

**Redraw rule for Gemini / Nano Banana:** keep every box, group and arrow listed. Do not add, merge or drop any. Styling is free; structure is fixed.

|                                                                                                                                                                                            |                                        |                                                 |                                             |                                 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|-------------------------------------------------|---------------------------------------------|---------------------------------|
| COMPETITOR TABLE (✓ = documented public evidence · ? = not found / unverified · (P) = planned by us, not built · n/a = not that kind of system)<br>Never read "?" as "they don't have it". |                                        |                                                 |                                             |                                 |
| Feature                                                                                                                                                                                    |                                        | Ours                                            | The Syndicate*                              | ARAI scenario repo   WMG Safety |
| Pool India   Swaayatt Robots                                                                                                                                                               |                                        | Minus Zero                                      |                                             |                                 |
| Indian traffic / scenario work                                                                                                                                                             |                                        | ✓ catalogue of 90;                              | (P) executable suite                        | ?   ✓   ✓                       |
| crowdsourced                                                                                                                                                                               | ✓ company-described                    | ✓ company-described                             |                                             |                                 |
| Hand-signal reading                                                                                                                                                                        |                                        | (P)                                             | ?                                           | ?                               |
| ?                                                                                                                                                                                          | ?                                      |                                                 |                                             |                                 |
| Reactive traffic (reacts to our car)                                                                                                                                                       |                                        | (P)                                             | ?                                           | ?                               |
| authoring tool could not yet model an actor slowing/restarting)                                                                                                                            |                                        | n/a                                             | n/a                                         | ? (paper:                       |
| Independent movement checker (no ML)                                                                                                                                                       |                                        | (P)                                             | ?                                           | n/a (not a planner)  n/a        |
| ?                                                                                                                                                                                          | ? (claims onboard "safety mechanisms") |                                                 |                                             |                                 |
| Paired Indian closed-loop metrics                                                                                                                                                          |                                        | (P)                                             | ?                                           | ?                               |
| ?                                                                                                                                                                                          | ?                                      |                                                 |                                             |                                 |
| MATLAB/Simulink end-to-end (PS toolchain)                                                                                                                                                  | (P)                                    | ?                                               | ?                                           | ?                               |
| ?                                                                                                                                                                                          | ?                                      |                                                 |                                             |                                 |
| Drives on real public roads                                                                                                                                                                |                                        | x simulation only                               | ?                                           | test track (separate facility)  |
| n/a                                                                                                                                                                                        | ✓ company-described                    | ✓ Bengaluru, safety driver, ≤25 km/h (May 2025) |                                             |                                 |
| *Teammate-supplied description only (stock Western CARLA, YOLOv11 + LiDAR, 3-tier planner, 3 easy scenes, no numbers shown); not independently verified – so no ticks or crosses.          |                                        |                                                 |                                             |                                 |
| Sources: DOC 3 R25–R28; P1 competitor check; P3.                                                                                                                                           |                                        |                                                 |                                             |                                 |
| CASH COST (what we actually pay)                                                                                                                                                           |                                        |                                                 |                                             |                                 |
| Item                                                                                                                                                                                       |                                        | Cost                                            | Basis                                       |                                 |
| MATLAB + Simulink + toolboxes (R2026a)                                                                                                                                                     |                                        | ₹0                                              | KIET campus licence                         |                                 |
| CARLA 0.9.15 · SUMO                                                                                                                                                                        |                                        | ₹0                                              | open source                                 |                                 |
| Indian datasets (IDD family, METEOR...)                                                                                                                                                    |                                        | ₹0                                              | academic use; some need sign-up / agreement |                                 |
| Training compute                                                                                                                                                                           |                                        | ₹0                                              | KIET DGX A100                               |                                 |
| Backup cloud A100 (if DGX busy)                                                                                                                                                            |                                        | ~\$0.43–1.09 per GPU-hour → 200 h ≈ \$86–218    |                                             |                                 |
| getdeploying.com/gpus/nvidia-a100, thundercompute.com (Sep 2026)                                                                                                                           |                                        |                                                 |                                             |                                 |
| Windows lab PC (8 GB GPU) for CARLA                                                                                                                                                        |                                        | ₹0                                              | existing KIET lab                           |                                 |
| Real dashcam clip                                                                                                                                                                          |                                        | ₹0                                              | phone camera                                |                                 |
| => Real cost is builder time + compute hours, not money. The pie shows the planned effort split (a plan, not a measurement).                                                               |                                        |                                                 |                                             |                                 |

## Mermaid source (paste into mermaid.live to re-render)

```
pie showData
title Planned effort split – 20 builder-weeks (2 builders × 10 weeks)
"Brain + safety checker (A W1–W3, W6)" : 4
"Control + driver screen (A W4)" : 1
"Prediction + belief + scorer (A W5, W7)" : 2
"Testing, metrics, baselines (A W8–W10)" : 3
"CARLA world + Indian assets (S W1–W3, W6–W7)" : 5
"Perception ML + real-video test (S W4–W5, W9)" : 3
"Weather/potholes + final package (S W8, W10)" : 2
```